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PROBABILISTIC DETECTION OF SYSTEMIC DISEASES USING DEEP LEARNING ON FINGERNAIL BIOMARKERS

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***Abstract* – Fingernails act as accessible biomarkers for systemic health, often revealing early signs of conditions such as cardiovascular disease and diabetes through subtle architectural changes. However, these indicators are frequently overlooked in standard diagnoses, and existing AI solutions often stop at classification without inferring underlying risks. This study aimed to develop a deep learning-based system that classifies nail features and utilizes Bayesian inference to calculate the probabilistic likelihood of systemic diseases. The researchers utilized a labeled dataset of 7,258 fingernail images across 10 distinct classes (e.g., Clubbing, Koilonychia, Terry's Nails), verified by a dermatologist. To bridge visual recognition with clinical diagnosis, a statistical dataset was curated from peer-reviewed literature to map nail features to systemic diseases using conditional probabilities. Five deep learning architectures—VGG-16, ResNet-50, EfficientNetV2-S, SwinV2-T, and ConvNeXt-Tiny—were trained using various transfer learning strategies, including full fine-tuning and gradual unfreezing. Performance evaluation demonstrated that modern architectures utilizing a Gradual Unfreezing strategy yielded the best results. ConvNeXt-Tiny achieved the highest accuracy at 88.93%, followed closely by the vision transformer SwinV2-T at 88.27%, significantly outperforming the VGG-16 baseline of 74.92%. The study successfully deployed these models into a web-based prototype that integrates user demographics (age and sex) with Bayesian inference to output ranked probabilities of systemic conditions, providing an interpretable and non-invasive health screening tool. The findings conclude that combining deep learning with probabilistic modeling enhances diagnostic utility by filtering implausible conditions and providing scientifically grounded risk assessments.**

Keywords: Deep Learning, Bayesian Inference, Fingernail Biomarkers, Systemic Diseases, Vision Transformers (ViT), Convolutional Neural Network (CNN), Preventive Healthcare

I. INTRODUCTION

Fingernails are often referred to as a "window to systemic health," (Singal & Arora, 2015) as they can reveal early signs of serious conditions such as diabetes, cardiovascular diseases, and liver disorders through subtle changes in their appearance. These abnormalities, such as Beau's lines (horizontal ridges indicating stress or illness), clubbing (enlarged fingertips linked to heart or lung issues), or pitting (small depressions associated with psoriasis or other systemic diseases), frequently appear before other symptoms become noticeable. Despite their diagnostic potential, these signs are commonly overlooked during routine medical checkups due to their subtle nature and the lack of specialized tools or training for general practitioners to identify them. This oversight delays early intervention, which could significantly improve health outcomes, particularly for individuals in underserved communities with limited access to advanced diagnostics.

The importance of accessible, non-invasive diagnostic methods cannot be overstated, as they empower individuals to monitor their health proactively and seek timely medical advice. However, many people worldwide face barriers to such healthcare services, including geographical isolation, financial constraints, and a lack of awareness about the significance of nail abnormalities. According to Gaurav et al. (2025), fingernails are a globally recognized source of biomarkers due to their visibility and ease of examination, yet their potential in preventive healthcare remains largely untapped. This gap highlights the urgent need for innovative solutions that can bridge these barriers and democratize early disease detection.

Artificial Intelligence (AI) has emerged as a transformative force in addressing such healthcare challenges, particularly through advancements in image processing and probabilistic modeling. Deep learning techniques, such as Convolutional Neural Networks (CNNs), excel at analyzing visual data, identifying patterns that may escape human observation. For example, a hybrid Capsule CNN achieved a 99.40% training accuracy in classifying nail disorders, showcasing the potential of deep learning in this domain (Shandilya et al., 2024). Similarly, a region-based CNN demonstrated superior performance to dermatologists in diagnosing onychomycosis, a common nail condition (Han et al., 2018). However, these studies often focus solely on classifying nail abnormalities without linking them to underlying systemic diseases, limiting their practical impact on preventive care.

In the Philippines, early efforts like the Bionyx project explored AI-driven fingernail analysis, using Microsoft Azure Custom Vision to identify systemic conditions such as heart, lung, and liver issues through nail images (Chua, 2018). While innovative, its reliance on older technology resulted in limited precision compared to modern deep learning models. Internationally, research has emphasized the diagnostic value of nails, with studies employing machine learning techniques like Support Vector Machines and CNNs to enhance classification accuracy (Dhanashree et al., 2022). Despite these advancements, a critical gap persists: the integration of deep learning-based classification with probabilistic inference to estimate the likelihood of systemic diseases, providing actionable insights for users.

To address the challenges faced in identifying systemic diseases from nail biomarkers, the study aims to develop a deep learning-based system that combines CNN models, specifically, EfficientNetV2-S, VGG-16, ResNet-50 and RegNetY-16GF. The study also utilized a newer vision transformer model, SwinV2-T for nail disorder classification with probabilistic models to infer systemic disease probabilities. By using publicly available datasets from Roboflow, the system is designed to be a globally accessible, non-invasive tool for early health screening. The proposed system will empower individuals, regardless of their location or socioeconomic status, to monitor their health proactively, offering a user-friendly platform that delivers probabilistic risk assessments and actionable recommendations for medical consultation.

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